

Lifelong Opportunity Guide 99 — Food Science Technician

Version: 2.0 controlled working master

Language: English

U.S. occupation benchmark: O*NET-SOC 19-4013.00 — Food Science Technicians

Canada comparison: NOC 22100 — Chemical technologists and technicians (Food technologist market title)

Colombia comparisons: CUOC 31160 — Técnicos en química industrial; CUOC 32573 includes Técnico de control calidad de procesamiento de alimentos

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What this career is

Food science technicians support the testing, quality, development, production, and safety work that helps food companies make consistent products and identify problems before they become larger risks.

Depending on the employer, a technician may work in a food laboratory, quality-control room, pilot plant, processing facility, production floor, research and development team, ingredient company, beverage plant, dairy facility, bakery, meat or seafood processor, produce operation, university laboratory, government laboratory, or contract-testing organization.

Typical job titles include food science technician, food laboratory technician, food quality technician, quality-control technician, laboratory assistant, quality-assurance technician, R&D technician, sensory technician, microbiology technician, sample technician, process-quality technician, or food technologist. Titles are not standardized. Always read the actual job description.

This guide uses **O*NET-SOC 19-4013.00 — Food Science Technicians** as the primary U.S. benchmark. The Canadian and Colombian occupations used here overlap with food laboratory, chemical-technology, and food-quality work but are broader in scope. They are comparisons, not exact one-to-one equivalents.

Why this role can be an opportunity

This field combines practical science with careful routine. It can suit people who enjoy:

- following a method step by step;
- measuring, weighing, timing, and recording accurately;

- noticing when a result looks unusual;
- working with samples, instruments, spreadsheets, and records;
- helping teams improve quality and consistency;
- understanding how ingredients and processes affect food;
- keeping work areas clean and organized;
- learning laboratory and production procedures;
- supporting food safety without pretending to hold authority they have not been assigned.

The occupation can also be demanding. Work may involve protective clothing, repetitive testing, strong cleaning odors, cold or warm processing areas, standing, shift schedules, strict sanitation rules, biological or chemical hazards, moving equipment, production deadlines, and the need to document errors rather than hide them.

A short laboratory course, food-processing course, supervised production role, quality-assurance trainee position, or job-shadowing experience can help a learner decide whether the environment fits before investing heavily in education.

Technician, technologist, scientist, and quality leader are different roles

Employers use titles differently, but authority matters.

A technician may be assigned to

- receive and label samples;
- prepare test materials;
- measure temperature, pH, moisture, weight, viscosity, color, concentration, texture, or other approved characteristics;
- prepare reagents or standards under procedure;
- operate approved laboratory instruments after training;
- perform sensory-support tasks under protocol;
- collect environmental or production samples;
- enter results into approved records or systems;
- compare results with established specifications;
- identify an out-of-range value;
- maintain equipment logs;
- clean and sanitize assigned tools or work areas;
- support traceability exercises;
- prepare retained samples;
- assist with pilot batches or product trials;
- report deviations to the authorized person.

A technologist or experienced specialist may additionally

- coordinate more complex testing;
- troubleshoot methods within assigned authority;
- train staff on approved procedures;
- coordinate laboratory schedules;
- support validation or verification work;
- review trends;
- help implement quality plans;
- support internal audits or investigations;
- contribute to process-improvement work.

A food scientist, authorized quality leader, preventive-controls qualified individual, laboratory signatory, engineer, regulatory specialist, or other accountable professional may have authority for

- approving or validating analytical methods;
- interpreting complex regulatory requirements;
- designing experiments or formulations at a professional level;
- approving food-safety plans;
- releasing or rejecting product;
- making final disposition decisions;
- signing regulated laboratory reports;
- determining corrective actions with legal or regulatory consequences;
- certifying compliance;
- communicating official conclusions to regulators or customers.

A technician should understand the process around these decisions without claiming authority that belongs to another role.

Typical responsibilities

Depending on the workplace, responsibilities can include:

- collecting samples according to an approved sampling plan;
- confirming sample identity, lot, date, time, location, and chain-of-custody information;
- preparing samples for physical, chemical, microbiological, or sensory analysis;
- following written standard operating procedures;
- measuring ingredients or test materials accurately;
- calibrating or verifying instruments when assigned and trained;
- recording calibration status and maintenance information;
- performing routine laboratory or in-process quality tests;
- monitoring production variables;

- checking packaging, labels, seals, weights, temperatures, or other approved quality attributes;
- supporting allergen-control or sanitation verification activities;
- documenting observations immediately and accurately;
- reporting out-of-specification or unusual results;
- maintaining laboratory notebooks, worksheets, electronic records, or quality-system entries;
- preparing retained samples and reference materials;
- helping prepare pilot batches;
- supporting shelf-life or stability studies;
- assisting with sensory panels under approved protocols;
- supporting traceability and mock-recall exercises;
- checking that required records are complete;
- helping maintain inventory of laboratory supplies;
- following chemical, biological, waste, sanitation, and PPE procedures;
- supporting corrective-action investigations by providing accurate facts and records;
- helping prepare data for authorized reviewers.

What you must not do without authority

Do not assume that a technician title authorizes you to:

- release a batch or finished product;
- change a specification because a result is inconvenient;
- alter a test method without approval;
- retest repeatedly until a preferred result appears unless the approved procedure explicitly permits it;
- delete, backdate, overwrite, fabricate, or selectively omit laboratory data;
- sign another person's initials or electronic approval;
- state that a product is legally compliant unless that conclusion is assigned to you and supported by the organization's process;
- diagnose a foodborne illness;
- decide that a contamination event is harmless without authorized review;
- approve a supplier, ingredient, allergen statement, regulatory label, recall decision, or food-safety plan outside your role;
- mix chemicals by guesswork;
- bypass guards, interlocks, sanitation controls, or PPE requirements;
- taste or smell an unknown sample to identify it;
- enter confidential formulas, customer data, credentials, unpublished results, or regulated records into an unapproved AI tool;

- copy test results from another sample or lot;
- conceal a deviation because production is behind schedule.

When in doubt, stop, preserve the evidence, and escalate.

The basic laboratory and quality workflow

A simple sequence can help prevent errors.

1. Confirm the assignment

Before testing, confirm:

- what sample or material you are handling;
- the correct method or SOP;
- required PPE;
- equipment needed;
- acceptance criteria or specification;
- who owns the final decision;
- how results must be recorded;
- what to do if something is abnormal.

2. Protect sample identity

Check labels, lot numbers, collection time, sample condition, container, storage requirements, and chain-of-custody information. A technically perfect test performed on the wrong sample is still a failed result.

3. Verify equipment status

Confirm the instrument is approved for use, within required calibration or verification status, clean, and appropriate for the method. If equipment status is uncertain, do not improvise.

4. Perform the method exactly

Follow the approved sequence. Record actual measurements and observations. Do not round, substitute, or modify values unless the method says how.

5. Review for obvious problems

Before submitting results, check sample identity, units, decimal placement, calculations, transcription, instrument messages, blanks, controls, standards, and required signatures or electronic fields.

6. Escalate deviations

If a result is out of specification, a control fails, a sample is damaged, a temperature excursion occurs, contamination is suspected, or data integrity

is questionable, preserve the record and use the employer's deviation process.

Food safety and sanitation

Food science technicians often work close to systems that protect consumers. The exact rules depend on food type, country, facility, and process.

In the United States, FDA's 21 CFR Part 117 framework covers current good manufacturing practices and, for covered facilities, hazard analysis and risk-based preventive controls for human food. CGMP topics include employee hygiene, sanitary operations, equipment, facility conditions, production/process controls, and allergen cross-contact prevention.

A technician should understand practical controls such as:

- hand hygiene;
- clean protective clothing;
- separation of raw and ready-to-eat materials where required;
- allergen-control procedures;
- sanitation and pre-operational checks;
- temperature control;
- correct chemical concentration;
- prevention of foreign-material contamination;
- controlled sample handling;
- environmental monitoring where assigned;
- avoiding cross-contamination between samples;
- documented corrective action and escalation.

Food safety is not a slogan. It is a system of defined responsibilities, documented controls, verified actions, and escalation when something goes wrong.

Allergen awareness

Food allergens can create serious risk. A technician may support allergen verification, label checks, sanitation validation, changeover checks, environmental swabbing, or ingredient-control activities.

Do not:

- assume that visually clean equipment is allergen-free;
- move allergen materials between areas without following procedure;
- substitute ingredients without authorized review;
- change allergen statements on labels;
- decide that a positive or questionable allergen result can be ignored.

Follow the approved program and escalate uncertainty.

Microbiology and biological hazards

Some food laboratories handle microorganisms or samples that may contain pathogens.

Only perform tasks for which you are trained and authorized. Follow the laboratory's biosafety, PPE, decontamination, waste, exposure, and incident procedures.

Important habits include:

- treat unknown cultures and environmental samples cautiously;
- keep clean and contaminated materials separated;
- disinfect according to validated procedures;
- do not take laboratory materials home;
- never eat or drink in prohibited areas;
- report spills, exposures, broken containers, or compromised samples immediately;
- do not reopen or manipulate cultures outside the approved method.

Chemicals and reagents

Food laboratories may use acids, bases, solvents, stains, sanitizers, standards, and other chemicals.

Before use:

- confirm identity and concentration;
- read the applicable label and safety information;
- use the specified PPE;
- use ventilation or a hood when required;
- check compatibility before mixing;
- use approved storage;
- label prepared solutions correctly;
- record preparation date, expiration, or preparer information when required;
- dispose of waste through approved procedures.

Never mix chemicals to “see what happens.”

Production and equipment safety

Technicians who collect samples or perform checks on production lines may work near conveyors, mixers, cutters, ovens, fryers, pressurized systems, refrigeration equipment, forklifts, hot surfaces, steam, blades, and cleaning operations.

Follow site rules for:

- machine guarding;
- lockout/tagout or hazardous-energy controls;
- authorized access areas;
- traffic routes;
- hearing protection;
- heat and cold exposure;
- slip and fall prevention;
- confined-space restrictions;
- chemical sanitation systems;
- sample ports and hot-process sampling.

A quality deadline never justifies bypassing a safety control.

Data integrity

Food-quality decisions can depend on a small number of measurements. That makes record integrity essential.

Good data practice means:

- record results when they are generated;
- use the correct sample identifier;
- preserve original observations;
- make corrections through the approved process;
- never erase or conceal the original value improperly;
- use your own account and credentials;
- never share electronic signatures;
- document repeated tests and reasons according to procedure;
- preserve instrument files when required;
- report suspected data manipulation.

If you make an honest error, report it. A corrected, traceable mistake is safer than a hidden one.

Cybersecurity and connected laboratories

Modern laboratories may use networked balances, chromatography systems, environmental-monitoring systems, laboratory information management systems, cloud applications, barcode systems, shared drives, and quality-management software.

Practical cybersecurity habits include:

- use employer-approved systems;
- enable multifactor authentication where supported;

- use unique passwords or an approved password manager;
- do not install unauthorized software on laboratory computers;
- verify unusual password-reset or payment requests;
- do not move regulated or proprietary data to personal email or personal cloud storage;
- lock screens when leaving a workstation;
- report suspicious files, phishing messages, or unexplained system behavior;
- do not connect personal USB devices unless policy permits it;
- preserve records if a system incident affects test data.

Cybersecurity is part of data integrity when systems contain test results, formulas, batch information, supplier records, or release data.

Responsible AI

AI may be helpful for low-risk learning and drafting when the organization permits it.

Reasonable uses can include:

- explaining public scientific concepts in simpler language;
- drafting a study outline;
- brainstorming practice interview questions;
- creating a fictional checklist for training;
- proofreading a non-confidential report draft;
- summarizing public regulatory guidance for later human verification.

Do not use AI as the final authority for:

- laboratory calculations that control release decisions;
- interpreting an out-of-specification result;
- changing a validated method;
- regulatory requirements;
- food-safety plans;
- allergen decisions;
- contamination investigations;
- recalls;
- product disposition;
- validation conclusions;
- medical or toxicological conclusions.

Do not place confidential formulas, process parameters, customer or supplier confidential information, test results, credentials, personal data, regulated records, or security-sensitive facility information into an unapproved AI system.

A useful rule is: **AI may help organize or explain; authorized humans and approved systems must verify consequential decisions.**

Skills employers value

Science and laboratory basics

- measurement and units;
- basic chemistry;
- basic microbiology;
- sample preparation;
- solution preparation;
- dilution concepts;
- pH;
- temperature;
- moisture;
- weight and mass;
- basic statistics;
- instrument care;
- contamination prevention.

Quality skills

- following SOPs;
- specification awareness;
- sampling discipline;
- traceability;
- lot control;
- document control;
- deviation reporting;
- change-control awareness;
- corrective-action awareness;
- calibration awareness;
- audit-ready records;
- attention to detail.

Digital skills

- spreadsheets;
- word processing;
- email and calendars;
- laboratory information systems;
- quality-management systems;
- barcode systems;
- approved cloud tools;
- data entry with unit and decimal accuracy.

Human skills

- reliability;
- clear communication;
- patience;
- willingness to ask questions;
- comfort reporting bad news;
- teamwork;
- respect for procedures;
- professional curiosity;
- ability to work under production pressure without falsifying data.

U.S. labour-market snapshot

O*NET-SOC **19-4013.00 — Food Science Technicians** is the primary U.S. benchmark for this guide.

Current O*NET/BLS-linked information reports:

- **2025 median hourly wage:** \$25.06;
- **2025 median annual wage:** \$52,130;
- **2025 10th percentile:** \$39,820/year;
- **2025 90th percentile:** \$78,820/year;
- **2024 employment:** about 20,400;
- **2024-2034 projected growth:** 5% to 6%;
- **projected openings reported by O*NET:** about 3,200 over the projection period.

BLS's Occupational Outlook Handbook separately reported a May 2024 median annual wage of **\$49,430** for food science technicians. The difference reflects data year and reporting source; do not treat it as a contradiction.

Pay varies by geography, employer, shift, industry, education, experience, laboratory responsibility, and technical specialization.

Current private-market estimate — United States

Private salary websites are not official government statistics.

ZipRecruiter reported an estimated U.S. average of about **\$46,976/year (\$22.58/hour)** for the market title Food Science Technician as of July 8, 2026, with a typical 25th-to-75th-percentile range around **\$34,000 to \$59,000/year**.

Glassdoor displayed a higher 2026 estimate using a relatively small self-reported sample. The disagreement is a reminder that private salary sites use different methods and sample sizes.

Use official BLS/O*NET data as the main benchmark and private estimates only as current market context.

Education and entry pathways — United States

There is no single universal entry credential for every food science technician job.

O*NET responses show employers using a mix of bachelor's, associate, and high-school-level entry expectations. BLS also notes that education depends on the specialty and assignment.

Relevant education can include:

- food science;
- food technology;
- chemistry;
- biology;
- microbiology;
- biotechnology;
- laboratory technology;
- quality assurance;
- food processing;
- manufacturing technology;
- culinary or food-production programs with strong safety and science content.

A learner who cannot immediately afford a four-year degree can still build evidence through a community-college course, laboratory certificate, quality role, production job, food-safety training, or employer-supported learning, then reassess after gaining experience.

Funding and low-cost training — United States

CareerOneStop and American Job Centers

CareerOneStop provides a WIOA-Eligible Training Program Finder:

[Source link](#)

Search more than one title. Programs may appear under food science, laboratory technology, chemistry, biotechnology, quality assurance, food processing, manufacturing, or industrial technology.

WIOA eligibility, approved providers, supportive services, and funding vary by state and individual circumstances. Funding is not automatic. An American Job Center can explain local options.

Community colleges

Community colleges often provide lower-cost science prerequisites, laboratory courses, certificates, and associate degrees. Ask about:

- adult learner grants;
- state workforce grants;
- employer tuition assistance;
- Pell Grant eligibility for qualifying programs;
- payment plans;
- credit for prior learning;
- laboratory assistant jobs on campus;
- cooperative education.

Work-based learning

Search Apprenticeship.gov, but do not assume a formal registered apprenticeship exists locally for this exact occupation.

Other practical routes may include:

- QA technician trainee;
- laboratory assistant;
- sample receiving technician;
- production quality inspector;
- sanitation verification role;
- paid internship;
- cooperative education;
- employer-sponsored GMP or HACCP training;
- entry production roles with internal promotion into quality.

Canada pathway

Canada's Job Bank uses the market title **Food technologist** under **NOC 22100 — Chemical technologists and technicians**. This group is broader than the U.S. benchmark.

Job Bank reports Canada-wide hourly wages, updated November 19, 2025, of:

- **low:** C\$21.00;
- **median:** C\$29.80;
- **high:** C\$47.69.

Typical Canadian requirements indicate:

- technologists usually complete a two- or three-year college program in chemical, biochemical, chemical engineering technology, or a closely related discipline;
- technicians usually complete a one- or two-year college program in a related field;
- certification may be available or requested depending on province, field, and employer.

Job prospects vary by province. Job Bank reports national labour demand and supply for NOC 22100 as broadly balanced over 2024–2033.

Canada training supports

Start with:

[Source link](#)

and current provincial or territorial employment/training services.

Support can include student aid, skills training, employment services, work experience, and programs funded through federal-provincial agreements. Eligibility and availability vary. Verify before enrolling.

Colombia pathway

Colombia offers relevant technical and technologist routes through both occupational classifications and SENA training.

Relevant OCUPACOL comparisons

CUOC 31160 — Técnicos en química industrial includes **Técnico químico de alimentos**.

CUOC 32573 includes **Técnico de control calidad de procesamiento de alimentos** among its occupational denominations.

These are useful comparators for food laboratory and quality-control work, not exact equivalents to U.S. SOC 19-4013.00.

SENA — Control de calidad en la industria de alimentos

SENA Betowa lists **Control de calidad en la industria de alimentos** as a **Tecnólogo** program with **3,984 hours**.

Competencies include:

- physicochemical analysis;
- sensory analysis;
- microbiological testing;
- food-quality control;

- laboratory coordination;
- production verification;
- safety and environmental practices;
- digital tools;
- practical training.

Program page:

[Source link](#)

SENA — Procesamiento y conservación de alimentos

This **Tecnólogo** pathway is also listed at **3,984 hours** and covers receiving, processing, preservation, hygienization, production, and quality verification.

[Source link](#)

SENA — Protección y conservación de alimentos

SENA also lists virtual learning that covers contaminants, good manufacturing practices, foodborne illness, and quality assurance:

[Source link](#)

SENA cohorts, seats, cities, dates, and modalities change. Verify the current Betowa listing before making plans.

Latin America and the Caribbean

For readers outside Colombia, OIT/Cinterfor can help identify national vocational-training institutions:

<https://www.oitcinterfor.org/>

<https://www.oitcinterfor.org/statsfp/paises>

Search each country's official training system for terms such as:

- tecnología de alimentos;
- ciencia de alimentos;
- control de calidad;
- laboratorio;
- inocuidad alimentaria;
- microbiología;
- procesamiento de alimentos;
- química aplicada;
- aseguramiento de calidad.

Verify program level, cost, modality, admission requirements, current seats, and recognition with the official institution.

Certifications and short courses

Some employers value training in topics such as:

- GMP or CGMP;
- HACCP;
- food allergen control;
- food defense;
- internal auditing;
- laboratory safety;
- ISO-based quality systems;
- sanitation;
- sensory testing;
- root-cause analysis;
- statistical process control.

Treat certifications carefully. A course certificate can show training, but it does not automatically make the holder a food-safety authority, regulatory expert, laboratory signatory, preventive-controls qualified individual, or accredited auditor.

Before paying for a credential, check real job postings in your target region and ask whether employers actually request it.

Free and low-cost learning sequence

A learner can start without buying an expensive program immediately.

Stage 1 — science refresh

Review:

- units and conversions;
- percentages;
- ratios;
- basic algebra;
- chemistry vocabulary;
- acids, bases, and pH;
- microbiology basics;
- temperature and measurement.

Stage 2 — quality fundamentals

Learn:

- why specifications exist;
- what an SOP is;

- sample identification;
- calibration concepts;
- traceability;
- deviations;
- corrective action;
- documentation integrity.

Stage 3 — food safety

Study:

- contamination routes;
- allergens;
- hygiene;
- sanitation;
- time and temperature control;
- cross-contact;
- preventive controls;
- escalation.

Stage 4 — spreadsheet and documentation skills

Practice:

- entering results;
- unit labels;
- formulas;
- simple charts;
- sorting and filtering;
- controlled templates;
- version awareness;
- proofreading.

Stage 5 — practical exposure

Look for supervised work in:

- food production;
- QA/QC;
- laboratory support;
- sample receiving;
- sanitation verification;
- ingredient handling;
- pilot-plant support.

Stage 6 — targeted credential

Only after reviewing local job requirements, choose the least expensive credible credential that closes a real gap.

Accessibility and disability inclusion

Food laboratories and production environments vary widely. Some jobs involve long standing, fine motor work, color discrimination, lifting, cold rooms, protective clothing, repetitive movements, noise, or strong odors. Other roles are more desk- or instrument-based.

Do not assume a person cannot do the work because of a disability. Essential duties and accommodations depend on the specific job and jurisdiction.

Potential workplace supports may include:

- adjustable work height;
- anti-fatigue mats;
- seated work for appropriate tasks;
- task lighting;
- magnification;
- screen readers or zoom;
- accessible digital forms;
- written step sequences;
- color-independent labels where feasible;
- hearing protection or communication accommodations;
- scheduling adjustments;
- ergonomic tools;
- reduced-distraction documentation space.

Safety-critical tasks still require appropriate risk assessment and employer procedures. This guide does not certify any workplace or accommodation as legally sufficient.

Dysgraphia-friendly work strategies

For learners who struggle with handwriting, spelling, sequencing, or working memory:

- ask whether electronic records are permitted;
- use approved templates;
- keep a short pre-test checklist;
- separate sample ID from result entry visually;
- read numbers back before submitting;
- use units on every practice worksheet;

- use speech-to-text for non-regulated drafting if permitted;
- proofread critical identifiers manually;
- avoid relying on memory for multi-step methods;
- keep personal learning notes separate from regulated records.

Never use accessibility tools in a way that bypasses laboratory data-integrity or security controls.

Building a beginner portfolio

Do not use confidential employer data. Build fictional or public-data examples.

A portfolio might contain:

- a fictional sample-receipt log;
- a simple pH study using household-safe materials;
- a mock calibration log;
- a temperature-monitoring spreadsheet;
- a fictional deviation report;
- a sanitation checklist;
- a mock traceability exercise;
- a short explanation of why data integrity matters;
- a chart showing repeated measurements;
- a one-page SOP for a harmless practice task.

Label all examples as training simulations.

Résumé language

Useful evidence-based phrases include:

- followed documented procedures and maintained accurate records;
- supported quality-control testing and sample tracking;
- used spreadsheets to record and review measurement data;
- followed sanitation, PPE, and safety requirements;
- identified deviations and escalated them through the correct process;
- maintained traceable documentation;
- supported production or laboratory teams under time-sensitive conditions;
- protected confidential information and used approved systems.

Do not claim HACCP authority, microbiological expertise, regulatory certification, or laboratory methods you have not actually performed.

Interview preparation

Expect questions such as:

- How do you prevent sample mix-ups?
- What would you do if a result is out of specification?
- How do you handle a mistake in a laboratory record?
- Why is calibration important?
- What would you do if production pressures you to ignore a failed check?
- How do you protect food from cross-contamination?
- How do you handle confidential formulas or test data?
- When should a technician stop work and escalate?

A strong answer usually shows procedure discipline, honesty, safety awareness, accurate documentation, and willingness to escalate.

Six-step action plan

This section intentionally repairs the legacy Guide 99 action-plan defect.

Step 1 — Choose a target environment

Select one realistic setting: food laboratory, QA/QC, beverage, dairy, bakery, meat/seafood, produce, ingredients, R&D, or processing quality.

Step 2 — Collect 10 real job descriptions

Record recurring requirements: education, shift, instruments, tests, GMP/HACCP, spreadsheets, lifting, microbiology, chemistry, sensory work, and experience.

Step 3 — Identify your three largest gaps

Choose only three. Example: basic chemistry, spreadsheet accuracy, and GMP knowledge.

Step 4 — Close one gap cheaply

Use community college, SENA, public workforce training, employer training, reputable free coursework, or a low-cost short course before buying a large credential.

Step 5 — Create evidence

Complete a supervised work task, internship, volunteer project, course laboratory, or fictional portfolio exercise that proves accuracy, documentation, and safety awareness.

Step 6 — Apply and review

Apply to technician, lab assistant, sample, quality, and trainee roles that match your current evidence. After every five to ten applications, review which requirements repeat and adjust training rather than collecting random certificates.

Before accepting a job

Ask practical questions:

- Which shift is this role?
- Is work mainly laboratory or production floor?
- What PPE is required?
- What chemicals or biological materials are handled?
- Is lifting required?
- Which tests and instruments are used?
- Who reviews out-of-specification results?
- Is weekend or holiday work common?
- What training is provided before independent testing?
- What documentation system is used?
- Is tuition assistance available?
- What is the progression from technician to senior technician or technologist?

Source and verification links

United States

- O*NET Food Science Technicians: [Source link](#)
- BLS Agricultural and Food Science Technicians: [Source link](#)
- BLS 2025 OEWS results: [Source link](#)
- CareerOneStop WIOA training finder: [Source link](#)
- Apprenticeship.gov: <https://www.apprenticeship.gov/>
- FDA CGMPs: [Source link](#)
- FDA Preventive Controls for Human Food: [Source link](#)
- FDA FSMA rules and guidance: [Source link](#)
- CISA Secure Our World: <https://www.cisa.gov/secure-our-world>
- NIST AI RMF: [Source link](#)

Canada

- Job Bank summary: [Source link](#)
- Job Bank wages: [Source link](#)
- Job Bank requirements: [Source link](#)
- Job Bank outlook: [Source link](#)

- Canada training gateway: [Source link](#)

Colombia and Latin America

- OCUPACOL CUOC 31160: [Source link](#)
- OCUPACOL CUOC 32573: [Source link](#)
- SENA Control de calidad en la industria de alimentos: [Source link](#)
- SENA Procesamiento y conservación de alimentos: [Source link](#)
- SENA Protección y conservación de alimentos: [Source link](#)
- OIT/Cinterfor: <https://www.oitcinterfor.org/>
- OIT/Cinterfor country resources:
<https://www.oitcinterfor.org/statsfp/paises>

Current non-government market context

- ZipRecruiter Food Science Technician salary: [Source link](#)
- Glassdoor Food Science Technician salary: [Source link](#)

Important notice

This guide is educational. It does not guarantee employment, income, admission, funding, apprenticeship placement, certification, regulatory compliance, food safety, laboratory competence, accessibility compliance, or any other outcome.

Requirements change by employer, food category, laboratory, jurisdiction, and time. Verify consequential decisions using current official sources, employer procedures, validated methods, and authorized professionals.

No independent human certification, certified translation, accreditation review, legal review, regulatory review, food-safety certification, or accessibility certification is claimed.

Author and AI assistance

Created and directed by **Alberto “Al” Leiva**. ChatGPT supported research, organization, editing, translation support, and document preparation under the author’s direction. The author remains responsible for editorial and publication decisions.

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